Design of Temperature Sensors based on Mach-Zehnder Interferometers for Environment & Health Applications

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***Index Terms*—Mach-Zehnder Interferometer, Silicon Photonics, Thermal Sensor**

I. Introduction

Among the current applications of integrated photonics, telecommunications remain the most prevalent, where photonic circuits transmit and guide information through waveguides and interferometers. However, with the advancement of silicon technology, emerging fields such as healthcare and environmental monitoring are gaining momentum. These applications often rely on evanescent wave sensing to measure specific fluid properties (e.g. concentration) by monitoring variations in the effective index. Such sensing mechanisms typically employ an asymmetric interferometer where one arm is exposed to the external environment while the reference arm remains encapsulated in oxide. Due to this inherent asymmetry, these sensors are highly sensitive to temperature fluctuations, which can lead to significant measurement errors. In this report, I propose the design of Mach-Zehnder Interferometer based temperature sensors, specifically aimed at decorrelating temperature-induced noise from the nominal sensor signal.

II. Theory

A straightforward method of a photonic integrated circuit relies on the confinement of light within a high-index core (Silicon, $n_{Si} \approx 3.47$) surrounded by a lower-index cladding (Silicon Dioxide, $n_{SiO_2} \approx 1.44$). The propagation of light is characterized by the effective index (n_{eff}), which represents the ratio between the speed of light in vacuum and the phase velocity in the waveguide.

Because of chromatic dispersion, n_{eff} depends on the wavelength λ (about 1550nm). To accurately describe the signal propagation in an imbalanced interferometer, we must use the group index (n_g), defined as $n_g = n_{eff} - \lambda * \frac{dn_{eff}}{d\lambda}$.

In this project, we consider two options : quasi-TM mode and quasi-TE mode since both have advantages. Quasi-TM mode maximizes the sensitivity of nominal sensors so coupling them with a temperature sensor implies to use also TM-mode for our thermal sensor. Quasi-TE mode implies less bending loss and is more relevant for temperature sensing since the effective thermo-optical coefficient (TOC_{eff}) is higher for silicon than for silicon oxide.

Considering a Mach-Zehnder Interferometer with two strip waveguides of different width at the middle, we can express the phase shift $\Delta\phi$ as following :

$$\Delta\phi = \frac{2\pi}{\lambda}(n_{eff,2}L_2 - n_{eff,1}L_1)$$

where $L_{1,2}$ is the length of arm 1 (reference) and arm 2 (sensitive). Since widths are different, $n_{eff,2}$ and $n_{eff,1}$ have not the same variations versus temperature. For each width to consider and each polarization, we must find the change in the waveguide effective index versus temperature. Assuming identical splitters (ideal 2x2 splitters in point symmetric configuration), the idealized power transmission (optical transfer function) $T(\lambda)$ is :

$$T(\lambda) = \frac{1}{2}(1 + \cos(\Delta\phi))$$

The free-spectral-range (FSR) is the wavelength spacing between two consecutive maxima of the transmission spectrum. A maximum occurs when the phase shift is equal to 2π. Thus, we can write $\frac{d\phi}{d\lambda} = \frac{2\pi}{FSR}$ and find an analytic formula for the FSR using the group index :

$$\frac{d\phi}{d\lambda} = -\frac{2\pi}{\lambda^2}\left(n_{eff,2}L_2 - \lambda L_2\frac{dn_{eff,2}}{d\lambda} - n_{eff,1}L_1 + \lambda L_1\frac{dn_{eff,1}}{d\lambda}\right)$$

$$FSR = \frac{\lambda^2}{n_{g,1}L_1 - n_{g,2}L_2} = \frac{\lambda^2}{n_{g,1}L_1 - n_{g,2}(L_1 + \Delta L)}$$

The temperature sensitivity S_T can be expressed as the resonance wavelength shift $d\lambda_c$ with respect to a little increase of temperature dT : $S_T = \frac{d\lambda_c}{dT}$. We neglect thermal expansion to consider $L_{1,2}$ as constants. As we analyze a peak shift, the corresponding phase shift still needs to be 2π : $\frac{d\Delta\phi}{dT} = 0$. This expression allows us to find S_T by first deriving phase shift with respect to temperature :

$$0 = 2\pi\left(\frac{d\left(\frac{1}{\lambda_c}\right)}{dT}(n_{eff,2}L_2 - n_{eff,1}L_1) + \frac{1}{\lambda_c}\left(\frac{dn_{eff,2}}{dT}L_2 - \frac{dn_{eff,1}}{dT}L_1\right)\right)$$

Taking into account thermo-optical effect and dispersion, we can write :

$$0 = -\frac{1}{\lambda_c^2}\frac{d\lambda_c}{dT}\left(n_{eff,2}L_2 - n_{eff,1}L_1\right) + \frac{1}{\lambda_c}\left(L_2\left(\frac{\delta n_{eff,2}}{\delta T} + \frac{\delta n_{eff,2}}{\delta \lambda}\frac{d\lambda_c}{dT}\right) - L_1\left(\frac{\delta n_{eff,1}}{\delta T} + \frac{\delta n_{eff,1}}{\delta \lambda}\frac{d\lambda_c}{dT}\right)\right)$$

Again, using the group index and isolating $\frac{d\lambda_c}{dT} = S_T$:

$$S_T = \lambda_c \frac{L_2\frac{dn_{eff,2}}{dT} - L_1\frac{dn_{eff,1}}{dT}}{n_{g2}L_2 - n_{g1}L_1}$$

For precise temperature sensing, the objective is to maximize S_T which means we need to have $L_2\frac{dn_{eff,2}}{dT} \gg L_1\frac{dn_{eff,1}}{dT}$ and $n_{g2}L_2 \approx n_{g1}L_1$. However, this last condition means we will have a huge FSR which is undesirable.

III. Modelling & Simulation

First, to choose the best widths for our MZI, we simulate a sweep from 350nm (to prevent excessive bending losses) to 600nm (to prevent multimode effects) with a 10nm step for a 220nm-high waveguide. For each width, we look at the group index (figure 1), the effective index (figure 2) the TE polarization fraction (figure 3) and the TOC_{eff} $\frac{\delta n_{eff}}{\delta T} = \frac{(n_{eff}(x1=0) - n_{eff}(x1=10))}{10-0}$ with x1 a Lumerical MODE parameter representing temperature (figure 4).

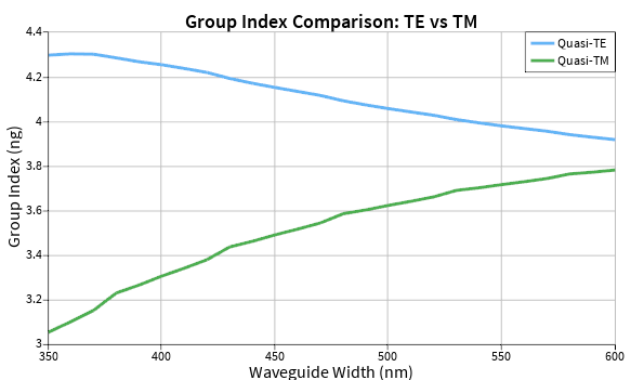

Fig. 1. Group index as function of the waveguide width for TE and TM polarizations

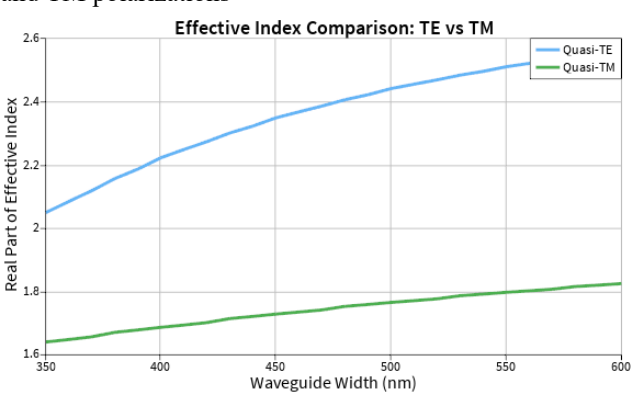

Fig. 2. Effective index as function of the waveguide width for TE and TM polarizations

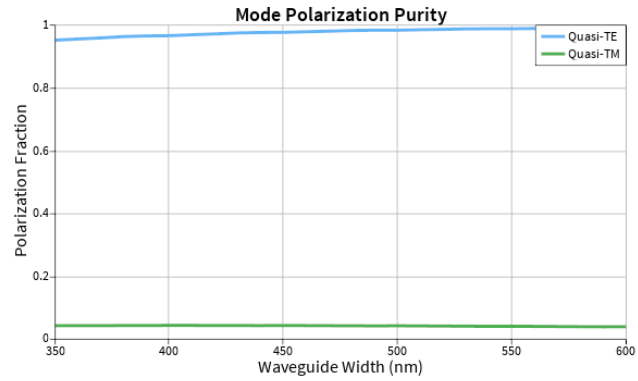

Fig. 3. Transverse polarization fraction (relatively constant with waveguide width for both TE and TM polarizations)

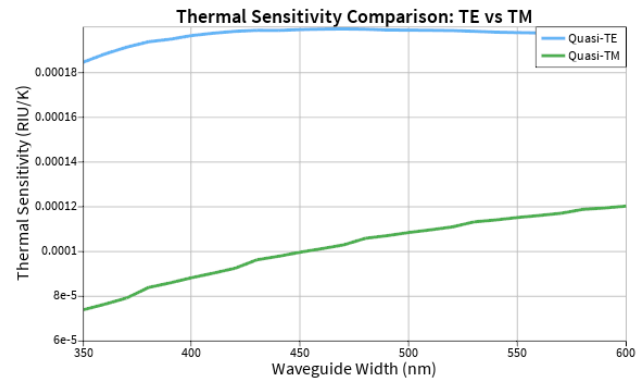

Fig. 3. Effective index thermal sensitivity as function of the waveguide width for TE and TM polarizations

These simulations confirm our expectations : thermal sensitivity and wavelength sensitivity (for effective index) are higher for quasi-TE mode than quasi-TM mode. Moreover, thermal sensitivity is maximum for a waveguide width of 470nm. For high MZI sensitivities, we need the TOC_{eff} to be different as much as possible for the two arms. Then, we also consider widths of 350nm, and 400nm. For quasi-TM mode, we also need to consider a width of 600nm which corresponds to the maximal thermal sensitivity.

Table 1 shows the different waveguide widths that will be used in our design and some parameters at $x1 = 0$ and quasi-TE mode. Table 2 shows the same for quasi-TM mode.

Waveguide	Width (nm)	TOC_{eff} (/K)	n_g (RIU)
A	350	0.000184	4.29581
B	400	0.000196	4.25296
C	470	0.000199	4.11549
D	600	0.000196	3.91756

Table 1. Design waveguides in quasi-TE mode

Waveguide	Width (nm)	TOC_{eff} (/K)	n_g (RIU)
E	350	0.000074	3.05538
F	400	0.000083	3.30655
G	470	0.000102	3.54477
H	600	0.000120	3.78180

Table 2. Design waveguides in quasi-TM mode

Thanks to these simulations, we are now able to calculate the length difference ΔL between the two arms in order to have a reasonable FSR of 10nm for MZI TE5 or TM5 in table 3 (same width). In this case, $\Delta L = \frac{\lambda^2}{FSR * n_g}$ so we calculate $\Delta L_{TE} = 58.4\mu m$ and $\Delta L_{TM} = 63.5\mu m$. For consistency, we choose a standard length difference for all MZIs of $60\mu m$. Then, to ensure an FSR of 15nm maximum for other designs, we can choose L_1, then L_2, so that all designs are respecting this condition.

$$L_1 = \frac{n_{g2}\Delta L + \frac{\lambda^2}{FSR}}{n_{g1} - n_{g2}}$$

We find $L_{1_{TE}} = 3000\mu m$ and $L_{1_{TM}} = 500\mu m$ which implies to use curvatures to have 10 designs in a 605 x 410 μm^2. We will be careful to use the same number of curvatures for the sensing arm and the reference one in order to prevent too high extinction ratio.

Finally, for the fabrication, we propose 10 different designs, summarized in table 3.

MZI	Waveguides 1-2	Expected FSR (nm)	Expected S_T (nm/K)
TE1	A-C	8.17	-0.297
TE2	A-C	8.17	-0.297
TE3	B-C	14.52	-0.195
TE4	D-C	2.86	0.038
TE5	C-C	9.73	0.075
TM1	E-H	4.07	0.080
TM2	E-H	4.07	0.080
TM3	F-H	5.17	0.077
TM4	G-H	6.96	0.071
TM5	H-H	10.59	0.049

Table 3. All designs to consider and expected results

Designs TE1 & TE2, and designs TM1 & TM2 are exactly the same in order to test manufacturing variability.

For all waveguides, figure 5 shows the electric field repartition in quasi-TE mode, and figure 6 for the quasi-TM mode. Figure 7 and figure 8 show respectively the effective index and group index versus wavelength.

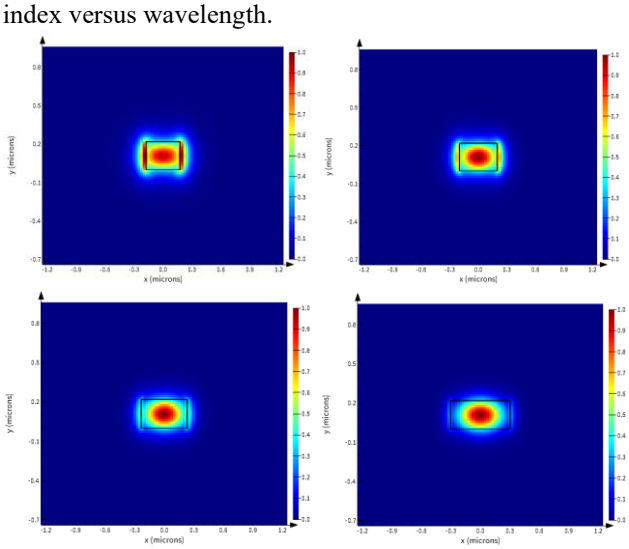

Fig. 5. Electric field repartition for waveguides A, B, C, D (in the sense of reading)

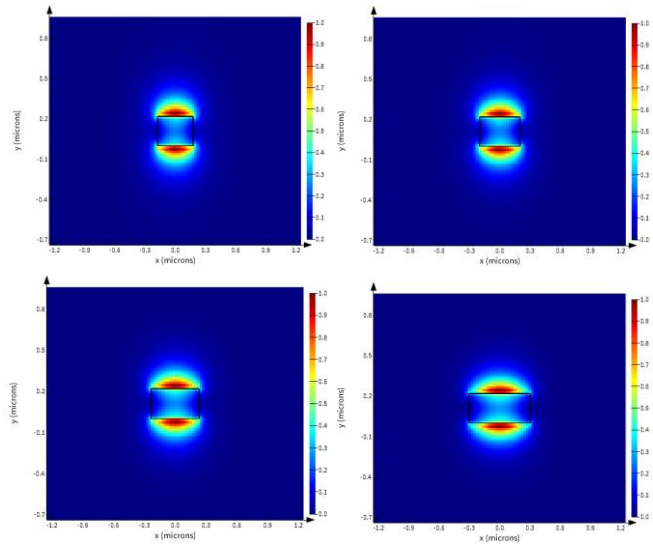

Fig. 6. Electric field repartition for waveguides E, F, G, H (in the sense of reading)

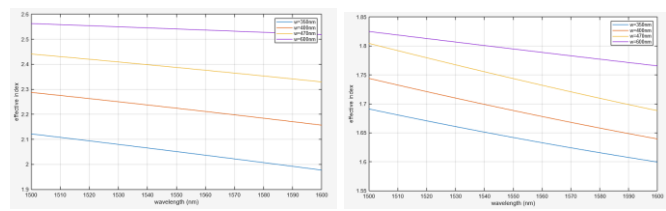

Fig. 7. Effective index variations versus wavelength for quasi-TE mode (on the left) and quasi-TM mode (on the right)

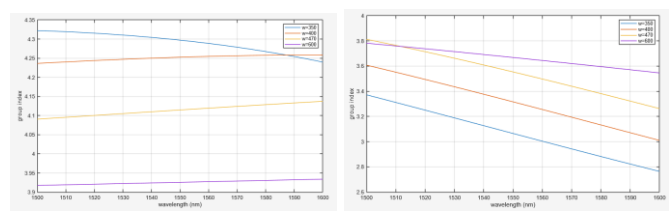

Fig. 8. Group index variations versus wavelength for quasi-TE mode (on the left) and quasi-TM mode (on the right)

Thus, we can define the compact model of the waveguides, which is a 2nd order polynomial that expresses effective index as a function of wavelength : $n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_c) + n_3(\lambda - \lambda_c)^2$. Coefficients n_1, n_2, and n_3 are given in table 4 for each waveguide width and polarization.

Waveguide	n_1	n_2	n_3
A	2.048865	-1.447772	0.260591
B	2.222480	-1.309214	-0.073050
C	2.385702	-1.115747	-0.151431
D	2.562761	-0.874124	-0.103503
E	1.640493	-0.915169	1.977047
F	1.686915	-1.045861	1.943035
G	1.741887	-1.162802	1.789043
H	1.825044	-1.263360	1.500454

Table 4. Compact model coefficients for all waveguides

For waveguide A, we need to be careful with bending losses as the width is small. For a typical bend radius of $5\mu m$, the mode-mismatch loss, simulated using the overlap calculations in Lumerical MODE, is $0.012dB$ for a 500nm-width waveguide and $0.022dB$ for a 350nm-width in quasi-TE mode. For quasi-TM mode, the same bend radius leads to a mode-mismatch loss of $0.396dB$ for a 350nm-width waveguide. We decided to decrease this value by increasing bend radius for quasi-TM mode designs (waveguides E to H) to $15\mu m$ (new mode-mismatch loss is $0.033dB$). Radiation losses are very small compared to mode-mismatch loss, and scattering losses (due to experiments : in the worst case $8dB/cm$) are calculated to be of the same order of magnitude as mode-mismatch loss. The total bend loss values are reasonable and are expected to have no such a huge influence on our measurements (if we have same numbers of bends for each arm of the MZI.

Figure 10 and Figure 11 shows the transmission spectrum of all designs without considering insertion loss in grating couplers and tapers to adjust arm widths (not available in Lumerical INTERCONNECT). Extracted FSR simulated values, shown in table 5, are near to the theoretical values reported in table 3. A clear non-linearity of the FSR occurs in designs TE1 and can be explained by a non-linear group index evolution versus wavelength, visible in figure 8. Figure 12 shows the effect of grating couplers insertion loss on the transmission spectrum of design TE1. It clearly demonstrates that insertion loss will be more important (6dB at minimum) than bend loss for less than a hundred curvatures per arm (order of magnitude). Figure 13 shows a graph on Lumerical INTERCONNECT representing typical photonic circuits for designs TE1.

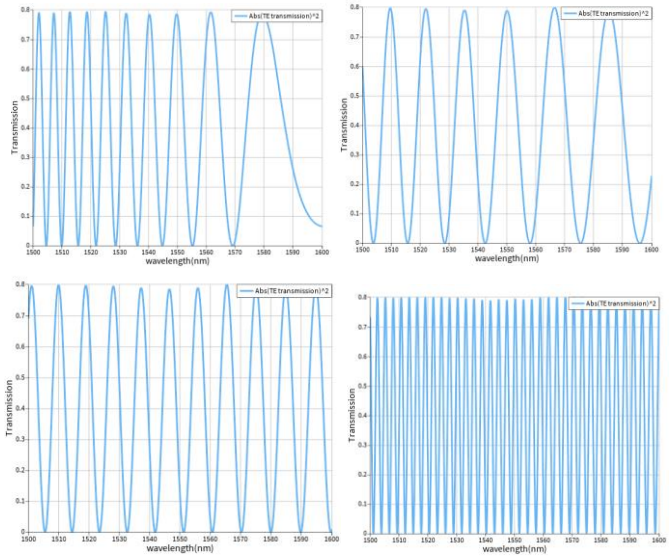

Fig. 10. Transmission spectra for MZIs TE1, TE3, TE4, and TE5 (in the sense of lecture)

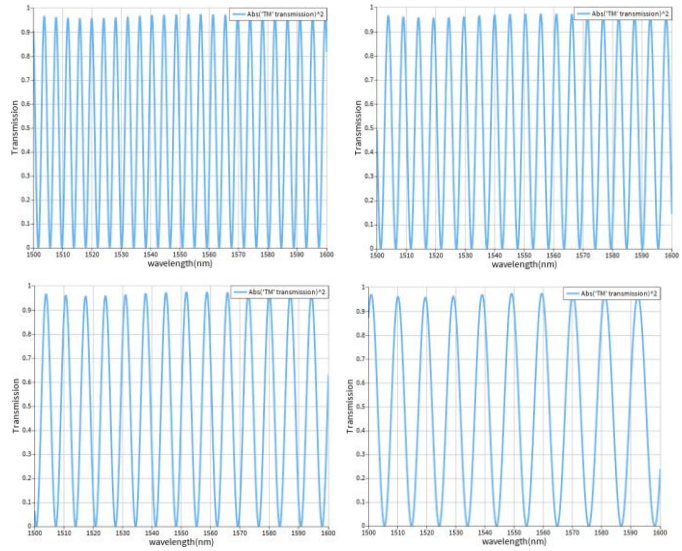

Fig. 11. Transmission spectra for MZIs TE1, TE3, TE4, and TE5 (in the sense of lecture)

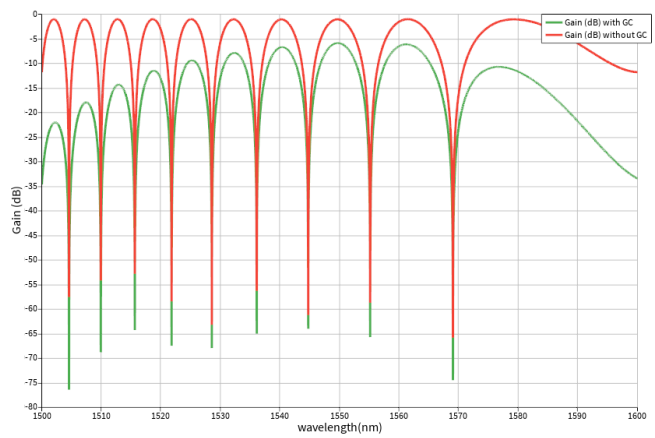

Fig. 12. Gain as function of wavelength for design TE1 with and without grating couplers in the photonic circuit

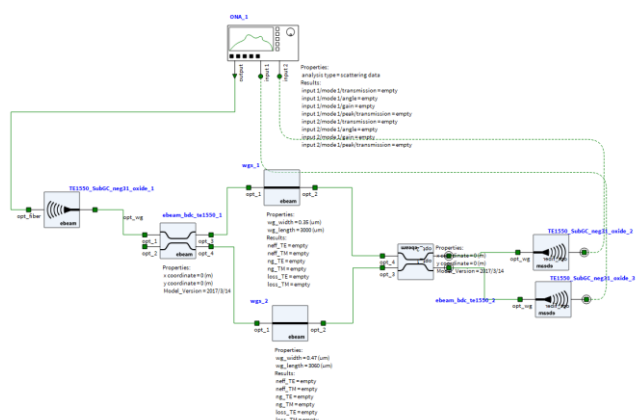

Fig. 13. Schema of a typical photonic circuit (grating couplers and MZI) on Lumerical INTERCONNECT

Finally, Lumerical INTERCONNECT can also simulate the temperature dependance of the circuit. By applying 25°C, 30°C, and 35°C, we can show the reflectance peak shift in figure 14. From these simulations, we extract a temperature sensitivity of

$-0.333nm/K$ which is not so far from the theoretical value calculated in table 3. The difference with theory is maybe due to precision in our sweeps, refractive index values (number of significant digits for TOC_{eff}), or reading errors due to FSR variations with wavelength near 1550nm. Reporting results for the same process applied to other designs in table 5, we see extremely high correlation with theoretical values (table 3) in terms of signs and values for FSR and S_T. It would be interesting to characterize thermally these sensors to confirm simulated and/or theoretical sensitivities. In this work, we will more focus on group index extraction from experimental data thanks to an FSR analysis.

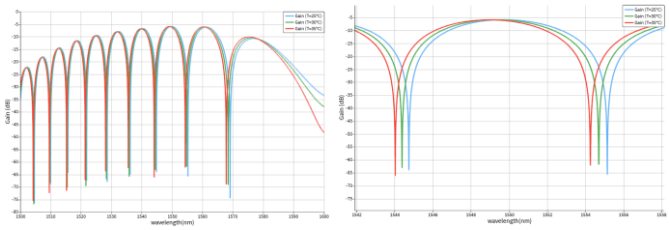

Fig. 14. Temperature influence on transmission spectrum for design TE1 (zoom on the right)

MZI	Simulated FSR (nm)*	Simulated S_T (nm/K)
TE1 & TE2	9.66	-0.333
TE3	15.49	-0.220
TE4	2.86	0.038
TE5	9.36	0.073
TM1 & TM2	4.16	0.081
TM3	5.24	0.087
TM4	6.96	0.072
TM5	10.23	0.049

Table 5. Simulated results for all designs. **Calculated near 1550nm. Also to precise that temperature coefficients are fixed for each width in agreement with table 1 & 2.*

IV. Fabrication

V. Experimental Data

VI. Analysis

VII. Conclusion

If our thermal measurements are limited by photodetector/laser precision (around 1pm), the best design we propose (TE1/2) is able to measure temperature differences as low as 3.33×10^{-4}°C which is remarkable for such a small device compared to literature [Sources]. For example, our device can measure a 100°C temperature difference by a 33nm-shift in the resonant wavelength near 1550nm. However, a disadvantage is the wavelength dependance if our laser is unstable : It would have been an idea to increase FSR and thus maybe also increasing again sensitivity, but it was not the purpose of this project.

We found good agreement between group index experimental values and theoretical ones for all designs. However, we 'sous-estimer' impact of loss due to bend and resulting phaseshift for asymmetric mach-zehnder interferometers.

To conclude, the cladding oxide presence allows to create a sensor sensitive only to temperature (not concentration for example and not pressure) which means we can successfully decorrelate parasite thermal signals in nominal sensors. This is true if the oxide is thick enough to consider no electric filed in the environmental fluid.

References